

Qatar-2b

R-0937

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180327 · created 2026-07-29T12:22:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458204.90102 +/- 0.00059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.012
Transit depth (Rp/Rs)^2	2.63 +/- 0.4 [%]
Orbital Inclination (inc)	85.84 +/- 1.44
Airmass coefficient 1 (a1)	0.962 +/- 0.012
Airmass coefficient 2 (a2)	0.027 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	#1 - [346, 53]
Optimal Method	PSF photometry
Transit Duration (day)	0.0692 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.162 ± 0.012 vs 0.16327 (deviation 0.11σ)
Duration fit vs published	0.0692 d vs 0.0767 d (deviation 1.29σ)
Mid-time O–C	-1.5 min
Depth SNR	13.5
Residual scatter	1.23 %

Light curve

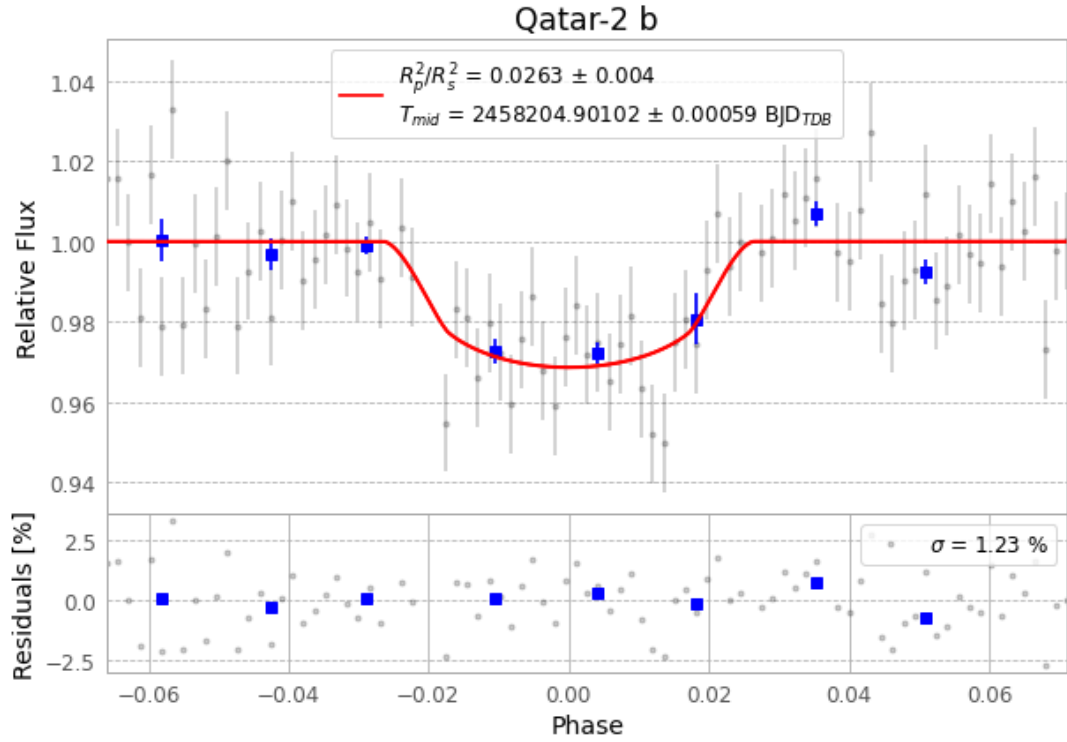


Figure 1 — FinalLightCurve_Qatar-2 b_2018-03-27.png (SHA-256 9f9e150c37b064b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [534, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6e653dc85f766610...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

87 FITS frames (57 MB), Qatar-2180327072112.FITS → Qatar-2180327114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180327.log (8c6c6d9143f1...), FinalLightCurve_Qatar-2 b_2018-03-27.png (9f9e150c37b0...), FinalLightCurve_Qatar-2 b_2018-03-27.csv (273d8defa39c...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 12:22Z.